A Scaling Reduction for Equality in the Multiplicative Infinitesimal Prekopa–Leindler Inequality

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Status

partial result / reduction. The result below does not characterize all geometric equality cases. It reduces the equality problem for the multiplicative inequality to the equality problem for the additive mean-form inequality, up to one scalar normalization.

Source Problem

The source paper is Sotiris Armeniakos and Jacopo Ulivelli, "A new infinitesimal form of the Prekopa–Leindler inequality with multiplicative structure and applications", arXiv:2602.10812. After proving Theorem 1.1 and Proposition 2.8, the authors state that a full characterization of equality cases for the multiplicative form is outside the paper and left for future work.

We conclude this section with a remark on the equality cases of Theorem 1.1. As anticipated in Theorem 1.2, $\langle\rho,\rho\rangle_{\mathrm{P}}=0$ if and only if $\rho\equiv 0$. Moreover, Livshyts [30] has proved that the equality cases of the Brascamp–Lieb inequality are saturated by constants when the inequality is restricted to compact sets. We shall now show that our new infinitesimal forms (27) and (5) of the Prékopa–Leindler inequality (1) admit further non-trivial equality cases. Observe that, by Lemma 2.7, equality in (27) implies equality in (5), but not vice-versa. That is, (5) is sharper than (27). A full characterization is out of the scope of this paper, and will be the subject of future research.

Proposition 2.8. *Consider K and μ as in Theorem 1.1. Then, for every $\alpha\geq 0, z\in\mathbb{R}$, and $x_0\in\mathbb{R}^n$, the functions $\rho=\alpha h_{K+x_0}(\nu_K)$ and $\varphi(x)=\alpha u^*(\nabla u(x-x_0))+z$ give equality in* (27) *and, therefore, in* (5).

Proof. From the equality cases of the Prékopa–Leindler inequality it is well known that, for every $w,v\in\mathrm{Conv}(\mathbb{R}^n)$ such that e^{-w} and e^{-v} are integrable,

$$\int_{\mathbb{R}^n} e^{-((1-t)\cdot w)\square(t\cdot v)}\,\mathrm{d}x=\left(\int_{\mathbb{R}^n} e^{-w}\,\mathrm{d}x\right)^{(1-t)}\left(\int_{\mathbb{R}^n} e^{-v}\,\mathrm{d}x\right)^{t} \tag{31}$$

if and only if the epigraphs of w and v are homothetic. That is, there exist $\alpha\geq 0, z\in\mathbb{R}$, and $x_0\in\mathbb{R}^n$ such that $v(x)=\alpha w(x-x_0)+z$. Suppose now that, in Theorem 2.6, $\rho=\alpha h_K(\nu_K)$ and $\varphi=\alpha u^*(\nabla u)$, In particular, $\psi=\alpha u^*$ in the concavity principle of (13). Recall that (27) is obtained from differentiating twice

$$\mathcal{S}(t)=\log\int_{\mathbb{R}^n} e^{-(u^*+t\alpha u^*)^*-I_{(1+t\alpha)K}}\,\mathrm{d}x.$$

By the properties of the Fenchel–Legendre transform and (12),

$$(u^*+t\alpha u^*)^*+I_{(1+t\alpha)K}=((1-t\alpha)\cdot u)\square(t\alpha\cdot u).$$

In particular, by (31), $\mathcal{S}(t)$ is linear and, therefore, $\mathcal{S}''(0)=0$, which implies equality in (27). In turn, by Lemma 2.7, equality is attained in (5). The complete statement is obtained considering arbitrary translations of h_K and u^*. □

3 Stability for the Poincaré inequality

The aim of this section is to prove Theorem 1.2. We point out that both the stability and the equality cases of (3) could be alternatively obtained by a careful inspection of its proof in [25] via the L_2 method. Here, instead, we propose a different approach based on Theorem 1.1. Further applications will be shown in the remaining sections.

The source defines three bilinear forms:

$$\begin{aligned}
\langle\rho_0,\rho_1\rangle_P &= \int_{\partial K}\langle \mathrm{II}^{-1}\nabla_{\partial K}\rho_0,\nabla_{\partial K}\rho_1\rangle\,d\mu-\int_{\partial K}H_\mu\rho_0\rho_1\,d\mu\\
&\quad+\frac{1}{\mu(K)}\left(\int_{\partial K}\rho_0\,d\mu\right)\left(\int_{\partial K}\rho_1\,d\mu\right),\\
\langle\varphi_0,\varphi_1\rangle_{BL} &= \int_K\langle(\nabla^2u)^{-1}\nabla\varphi_0,\nabla\varphi_1\rangle\,d\mu-\int_K\varphi_0\varphi_1\,d\mu\\
&\quad+\frac{1}{\mu(K)}\left(\int_K\varphi_0\,d\mu\right)\left(\int_K\varphi_1\,d\mu\right),\\
\langle\rho,\varphi\rangle_I &= \int_{\partial K}\rho\varphi\,d\mu-\frac{1}{\mu(K)}\left(\int_{\partial K}\rho\,d\mu\right)\left(\int_K\varphi\,d\mu\right).
\end{aligned}$$

The two inequalities are

$$\langle \rho, \varphi \rangle_I \le \frac{\langle \rho, \rho \rangle_P + \langle \varphi, \varphi \rangle_{BL}}{2}, \tag{M}$$

$$\langle \rho, \varphi \rangle_I^2 \le \langle \rho, \rho \rangle_P \langle \varphi, \varphi \rangle_{BL}. \tag{P}$$

Proof Intuition

The multiplicative inequality is obtained in the source by applying an abstract quadratic discriminant argument to the mean inequality. The same quadratic keeps exact equality information. For fixed ρ, φ, apply the mean inequality to $t\rho$ and φ. This gives a nonnegative quadratic in t. Multiplicative equality says precisely that this quadratic has zero discriminant, hence that it touches zero at one scalar normalization of ρ. That touching point is exactly equality in the mean inequality.

Theorem 1 (Scaling reduction for product equality)**.** *Let K, u, μ satisfy the hypotheses of Theorem 1.1 of arXiv:2602.10812, and let ρ and φ be admissible test functions. Put*

$$A = \langle \rho, \rho \rangle_P, \qquad B = \langle \varphi, \varphi \rangle_{BL}, \qquad C = \langle \rho, \varphi \rangle_I.$$

Assume $A > 0$ and $B > 0$. Then equality holds in the multiplicative inequality (P) for (ρ, φ) if and only if equality holds in the mean inequality (M) for $(\lambda\rho, \varphi)$, where

$$\lambda = \frac{C}{A}.$$

Equivalently, every nontrivial product-equality pair is a scalar re-normalization of a mean-equality pair, and conversely.

Proof. Apply (M) to $t\rho$ and φ, for arbitrary $t \in \mathbb{R}$. By bilinearity and homogeneity, this is

$$tC \le \frac{t^2 A + B}{2},$$

or

$$q(t) := At^2 - 2Ct + B \ge 0 \qquad \text{for every } t \in \mathbb{R}.$$

Since $A > 0$, the quadratic q is nonnegative for all t.

If equality holds in (P), then $C^2 = AB$. Hence

$$q(C/A) = A(C/A)^2 - 2C(C/A) + B = 0.$$

Thus equality holds in (M) for $(\lambda\rho, \varphi)$ with $\lambda = C/A$.

Conversely, suppose equality holds in (M) for $(\lambda\rho, \varphi)$ for some real λ. Then $q(\lambda) = 0$. Because $q(t) \ge 0$ for all t and $A > 0$, a real zero forces the discriminant to vanish:

$$(-2C)^2 - 4AB = 0.$$

Therefore $C^2 = AB$, which is equality in (P). □

Remark 1 (Trivial equality cases)**.** *The source records that $\langle \rho, \rho \rangle_P = 0$ if and only if $\rho \equiv 0$, and the Brascamp–Lieb kernel on compact sets consists of constants. Thus $\rho = 0$ or constant φ gives the trivial equality $0 = 0$ in (P). The theorem above isolates all nontrivial cases.*

Corollary 1 (Two-scale product equality family)**.** *Let (ρ_0, φ_0) be any nontrivial equality pair for the mean inequality (M); in particular, one may take the homothetic Prekopa–Leindler family supplied by Proposition 2.8 of the source paper:*

$$\rho_0 = \alpha h_{K+x_0}(\nu_K), \qquad \varphi_0(x) = \alpha u^*(\nabla u(x - x_0)) + z.$$

Then for every real s, t, and after adding an arbitrary constant to the second component if desired, the pair

$$(s\rho_0, t\varphi_0)$$

is an equality case for the multiplicative inequality (P). If (ρ_0, φ_0) is nontrivial, this rescaled pair is an equality case for the mean inequality (M) precisely when $s = t$.

Proof. The first claim follows from the homogeneity

$$\langle s\rho_0, t\varphi_0\rangle_I^2 = s^2t^2\langle \rho_0, \varphi_0\rangle_I^2 = s^2t^2\langle \rho_0, \rho_0\rangle_P\langle \varphi_0, \varphi_0\rangle_{BL}.$$

This is exactly

$$\langle s\rho_0, s\rho_0\rangle_P \, \langle t\varphi_0, t\varphi_0\rangle_{BL}$$

by homogeneity of the two quadratic forms. Constants may be added to φ_0 because both $\langle \cdot, \cdot\rangle_{BL}$ and $\langle \cdot, \cdot\rangle_I$ vanish on the constant part of the second variable.

For the final assertion, a nontrivial mean equality pair satisfies

$$\langle \rho_0, \varphi_0\rangle_I = \langle \rho_0, \rho_0\rangle_P = \langle \varphi_0, \varphi_0\rangle_{BL} > 0$$

by the arithmetic–geometric mean equality condition. Therefore mean equality for $(s\rho_0, t\varphi_0)$ is equivalent to

$$st = \frac{s^2 + t^2}{2},$$

which is $(s - t)^2 = 0$. $\square$

Limitations and Novelty Check

This is a reduction, not the full equality-case classification. It shows that future work on the multiplicative equality cases can focus on the mean-form equality cases: once those are known, all product equality cases follow by the single scalar de-normalization above, together with the trivial zero/constant-kernel cases.

The cheap run indexes had no prior entry for arXiv:2602.10812. A web search for the exact title and equality-case phrases found no later equality-case resolution. A same-author follow-up, arXiv:2602.23350, treats a strengthened dimensional Brunn–Minkowski condition and the (B)-conjecture, not the equality classification considered here.

References

References

[1] S. Armeniakos and J. Ulivelli, *A new infinitesimal form of the Prekopa–Leindler inequality with multiplicative structure and applications*, arXiv:2602.10812.

[2] S. Armeniakos and J. Ulivelli, *A strengthening of the dimensional Brunn–Minkowski conjecture implies the (B)-conjecture*, arXiv:2602.23350.